Research Square

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Exploring the Relationship Between the Flow of Visual Representations of Ideas and Designers' Creativity

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Abstract

This study aims to explore how do prototypes, sketches and CAD models supported designers during the early idea, idea development and finalization stages in the engineering design process flow. Using mixed-methods study, designer´s roles and contributions were evaluated with respect to the achieved design quality. 81 Professional industrial designers were observed and interviewed after participating in five workshop sessions. After exploring different permutations of sketching, prototyping and CAD modelling, results indicate that starting with “Prototyping” in the early idea stage contributed significantly to more ideas of higher quality, broadening the ideation space. Using CAD-modelling too early in the engineering design process constrain ideation and prematurely limit solution spaces.

Introduction

This study was initiated from designers´ needs to find the most creative engineering solutions in the early idea stages and convert these into feasible design concepts (Efimov-Soini, 2019; Tsai & Yang, 2017). It also evaluates three commonly used visual representation of ideas , namely prototyping, sketching, and computer-aided design (CAD) modelling, in terms of their respective representation qualities to support different phases of the design process (Howard, Culley, and Dekoninck, 2008; Tsai & Yang, 2017).

Recognised as formal design representations, prototypes, sketches and CAD-models have been accredited as aids for facilitating design research and ideation (Babapour, 2015; Camere & Bordegoni, 2016; Joundi et al., 2020; Kelly & Gero, 2014). Before 3D CAD models, physical prototypes and sketches were the most practical and fastest ways to convey and develop ideas for designers (Charlesworth, 2007; Tsai & Yang, 2017; Babapour, 2015). The introduction of 3D computer modelling software notably replaced, to some extent, traditional visualisation methods, which transformed the difficult, slow, and dirty creation process into a more efficient and clean virtual way of designing and prototyping (Isa & Liem, 2014). However, Charlesworth (2007) and Ekströmer & Wever (2019) suggested that designers may encounter restrictions when using CAD models during the early idea. Indeed, the use of CAD models and prototypes as design tools has been widely discussed concerning how they contributed to the thinking processes, as well as how the design activities were managed (Hallgrimsson, 2012 and Isa & Liem, 2020). On the one hand, several researchers discouraged using prototypes in early idea, because building prototypes early can be expensive and time-consuming and should therefore only be done when needed (Dow et al., 2010; Stowe, 2009; Viswanathan, Atilola, Esposito & Linsey, 2014; Murphy et. al, 2022). On the other hand, other researchers insisted that prototyping should be applied early in design and development activities to better understand problems, encourage faster learning, and increase communication among stakeholders and designers (Charlesworth, 2007; Isa & Liem, 2020; Jensen, Elverum, & Steinert, 2017; Leifer & Steinert, 2012; Sanders & Stappers, 2014).

However, limited research is available on the effectiveness of prototypes, sketches and CAD-models in the early generation of ideas (Hess & Summers, 2013; Babapour, 2015; Tsai & Yang, 2017). There is therefore a need for more research on the progression and transition of representations to explore and experience

different design activities experienced by designers during the early stages of the engineering design process. We draw on existing studies by Babapour, Hess, Howard and their associates to designate the flow of design activities to guide creativity as “early idea”, “idea development”, “and finalization” (Babapour, 2015; Camere & Bordegoni, 2016; Hess & Summers, 2013; Howard, Culley, and Dekoninck, 2008). This study addresses one research question: How do prototypes, sketches and CAD models supported designers during the early idea, idea development and finalization stages in the design process?

1. Creativity in the Engineering Design Process

This section introduces the early stages of the engineering design process, how design activities, i.e. prototyping, sketching and CAD-modelling are used to facilitate early-stage analysis and creative development, as well as how designers manage creativity constraints and design fixation.

1.1 Divergence and convergence steps in early stages in the Engineering Design Process-

According to a comprehensive literature study of 23 engineering design process frameworks (see Howard, Culley, & Dekoninck, 2008), six essential stages can be distinguished in a creative design process. These comprise *(1) establishing a need*, *(2) early idea development* (task analysis), *(3) idea development* (conceptual design), *(4) embodiment design*, *(5) final idea development* (detailed design), and *(6) implementation*. This study focuses on the representation aspects of the process: stages, (3), (4) and (5). During the “idea development” stage, designers iteratively explored the problem space to understand the context that the product will be a part of. Hereby, design problems were addressed holistically in a structured or unstructured manner through a process of transformative idea generation.

In the “embodiment design” stage, these initial ideas were then further developed, refined and more frequently verified against design requirements while addressing localised-, /sub-problems. During this stage, quantity was emphasised to generate as many solutions as possible. Especially, during the initial divergent phases, the problem-space was continuously evaluated and redefined from different perspectives so as to pre-empt too early fixation (Council, 2005; Gray, Brown & Macanufo, 2010; Liu et al., 2003; Meinel, Leifer & Plattner, 2011; Cross, 2021). Dym et al. (2005) defined the divergent process as an activity, where questions dominate facts to enlarge the creative space.

During the final ideation phase, mechanisms, components, user interfaces, aesthetics, and ergonomics of the product were further detailed once the final concept has been selected. Every component was designed according to the most suitable production technologies, with detailed specifications on the materials, sizes, and standard requirements (Cross, 2021)

The engineering design process is characterised as convergent, with alternating divergent and convergent moves, focussing on quantity at the beginning of the process, and consecutively narrowing down to

potential and a final solution, as illustrated in Figure 1 (Cropley, 2010; Howard et al., 2008; Leifer & Steinert, 2012; Liu, Chakrabarti & Bligh, 2003; Leifer and Steinert; 2012).

For divergent phases, the argument for valuing quantity over quality is based on studies showing that designers are more likely to acknowledge the uncertainties of the “so-called unknown knowns”, the more they explore and experiment (Bernal, Haymaker & Eastman, 2015; Jensen et al., 2017; Sutcliffe & Sawyer, 2013). For convergent phases, the contrary argument applies, where the possibilities were narrowed down and future actions are chosen (de Oliveira, Rozenfeld, Phaal & Probert, 2014). Dym et al. (2005) defined the convergent processes as an operating knowledge domain. In convergent processes, knowledge holds “true” values and is used to specify the actual task or refine and finalize the concept/product (Cropley, 2010).

1.2 Design Activities during early stages of the engineering design process

During the design process, several methods can be utilized to define the future requirements of the final product (Howard et al., 2008; Sutcliffe & Sawyer, 2013). In this paper, we focus on prototyping, sketching, and CAD modelling because they are among the most commonly used design representation tools to communicate design ideas during the concept development stage (Babapour, 2015; Tsai & Yang, 2017; Menold, et. al, 2019). Each of these design activities is briefly described below.

1.2.1 Sketching

As one of the most significant activities during the design process, sketching refers to producing visual images to externalise the process of visual thinking and assist in the creation and development of visual ideas (Mao et. al, 2020 and Ibrahim & Pour Rahimian, 2010; Bao, Faas, & Yang, 2018). Sketches signify a variety of information and are used in diverse ways (Babapour, 2015). A sketch is a preliminary and unpolished visual representation of something without emphasising details. More importantly, it is usually executed rapidly and presents only the key elements of the design. Sketches comprise informal freehand marks without the use of instruments and may include draft lines, text, dimensions, and calculations that help explain the meaning, context, and scale of the design (Pei, Campbell & Evans, 2011 and Bao, Faas, & Yang, 2018). The ambiguity of sketches distinctively creates assumptions, imaginations and reinterpretations, especially of those elements and views that are not explicitly shown. In other words, by omitting details in sketches, the creative space for follow-up ideas will be enlarged (Menezes & Lawson, 2006). However, Schütze, Sachse, & Römer (2003) and Bao, Faas, & Yang, (2018) suggested that sketches have a positive impact on the quality of the designed solution and facilitate individual experience during the designing process. As such, sketches can be perceived as a valuable tool for facilitating visual interactions among stakeholders in the early idea stages, because of their ambiguity (Menezes & Lawson, 2006; van der Lugt, 2005). According to the taxonomy by Pei at al. (2011), a diversity of visual representations can be distinguished for developing industrial and engineering design work.

1.2.2 Computer-Aided Design (CAD) Modelling

CAD modelling and other forms of digital modeling can be used to define the geometry and visual appearance as well as to facilitate simulations, variations, and appearance testing of future products (Tovey & Owen, 2000). Since their usage became more prominent in the early 2000s, it has been discussed whether they should be applied for only 2-D or to be extended to 3-D representation, especially in early-stage concept development (Ibrahim & Pour Rahimian, 2010; Ekströmer & Wever, 2019; Coyne et al. , 2002). Furthermore, studies conducted by Ibrahim & Pour Rahimian (2010) and Robertson and Radcliffe (2009) claimed that CAD should not be used during the early stages of product development as it can constraint exploration. Other studies present counter arguments outlining major advantages typically ascribed to CAD, such as improved solutions through broader explorations of the solution space as well as cost reduction in project work because of frequent and early iterations (Fixson and Marion, 2012). CAD has therefore been included in this study to further explore these tensions in conjunction with the design process.

1.2.3 Physical Prototyping

The progression in the construction of a physical representation of an idea is called physical prototyping (Houde & Hill, 1997; Lim, Stolterman & Tenenberg, 2008). According to Häggman, Tsai, Elsen, Honda, and Yang, (2015); Menold, et. Al, 2019), Neeley, Lim, Zhu, and Yang, (2013), prototyped designs are generated more quickly than those created using sketches or CAD. However, depending on their purpose, manifestations can be made through the use of different materials, and throughout the engineering design process, to facilitate exploration, evaluation, and communication (Beaudouin-Lafon & Mackay, 2007; Camere & Bordegoni, 2016; Houde & Hill, 1997; Jensen, Balters & Steinert, 2015; Isa and Liem, 2020; Murphy et. al, 2022).

During the initial stages of the engineering design process, low-fidelity physical prototypes can be produced from all types of materials and consistencies, including cardboard, foam, or wood, as well as through role-playing to facilitate exploration and defining requirements rather than concept testing (Ulrich & Eppinger, 2008; Yang, 2005). This indicates that a prototype shares a low resemblance with the final product (Sanders & Stappers, 2014; Yang, 2005), allowing accurately scaled and high-fidelity physical models to be constructed in the later stages for hypothesis and requirement testing. (Jensen et al., 2017; Sutcliffe & Sawyer, 2013; Ulrich & Eppinger, 2008).

1.3.Creativity Constraints and Design Fixation during the Design Process

The overall aim of the design process is in the end to generate a well thought-out solution containing some degree of creativity (Howard et al., 2008). To reach this goal, designers consciously guide themselves through divergent and convergent processes to develop the most creative and innovative solutions (Liu et al., 2003). However, creativity constraints and design fixation can affect the process (Crilly & Cardoso, 2017; Jensen, Birkeland & Steinert, 2016; Onarheim, 2012). Creativity constraints stem from the theory of priming effects which argues that the tightness of these constraints determines a designer´s creative space (Colombo, Bartesaghi, Simonelli & Antonietti, 2015; Onarheim, 2012). In this context, the professional expectations to use certain design tools, such as sketches, prototypes, or CAD-

models, can be perceived as creatively constraining or inspiring. How constraining or inspiring certain tools are, is dependent upon the number of ideas generated or the functionalities the participants end up with during prototyping sessions (Friedman, Fishbach, Förster & Werth, 2003; Onarheim, 2012). Nevertheless, Vasantha, Chakrabarti, Rout, and Corney, (2014) mentioned that irrespective of the design tool used, novice designers generated a lower number of redesigned concepts. Therefore, designers might require training for reinterpreting and extracting the necessary information from originally captured concepts, rather than working with poor understanding, ambiguity, and assumptions about the original designer's intent. This requires designers to balance a fine line between being constrained and inspired to enter the so-called sweet spot of creativity (Onarheim and Biskjaer,2015). To avoid over-constraining and under-constraining, Crilly and Cardoso (2017) described the concept of 'design fixation' as 'a blind adherence to a set of ideas or concepts limiting the output of conceptual design', which designers can mitigate by applying different methods. However, no specific claims were made on how prototypes, sketches, and CAD models support or undermine creativity, because of other underlying motives during different idea generation stages.

Methods And Procedure

Mixed-methods study were conducted based on identifying "Design Stage" and "Design Activity" as independent variables (Creswell, 2014; Chih-Pei, & Chang, 2017). The research methodology was established around a workshop approach, where participants were subjected to five consecutive sessions. Furthermore, qualitative data were gathered through design diaries, observations, and finally one-on-one interviews to provide insights on the effectiveness of the different design activities during creative explorations. This section describes how the procedures were applied in the study.

2.1 Mixed methods research experiments

To gain in-depth knowledge regarding the experiences of designers during early conception and development, mixed-method research experiments were used (Creswell, 2014; Chih-Pei, & Chang, 2017). This includes quantitative testing combined with qualitative observations. In this study, a comprehensive data-gathering exercise was conducted on a single individual, program, or event, to learn more about an unknown or poorly understood situation (Cash, Elias, Dekoninck & Culley, 2012). An experimental design approach was found most suited to approach the research question because it involved in-depth studies on the discovery and structuring of methodologies, the tools and techniques used in the design processes, and the exploration of design contexts to search for new product ideas.

To investigate how prototypes, sketches and CAD models supported designers during the early design process, a mixed method was developed using independent variables, i.e., the design activity and design phase, as well as dependent variables, including the numbers of sketches, ideas and prototypes. The design activity variables were "sketching", "CAD-modelling" and "prototyping", whereas the design stage variables were "early idea", "idea development", and "refinement" (Cash et al., 2012). After experimenting,

an ANOVA tests and Tukey test of Honest Significant Difference (Tukey's HSD) test were conducted in SPSS to answer the research question. The design experiment method developed in this study was adapted from Charlesworth (2007), Gioia, Corley, and Hamilton (2013), and Goldschmidt and Rodgers (2013).

For the qualitative observations, we relied on audio and video recordings, face-to-face interviews and designer diaries. Facilitators kept observation notes and took pictures during the entire design process. Expert reviews were used to evaluate the ideas. The data was analysed by reviewing audio and video recordings, field notes, and photographs taken during the design activities. Face-to-face interviews complemented with designer diaries were used to study the participants' feelings regarding the overall workshop, as well as to gain deeper, personal, and more valuable information (Seidman, 2006). These face-to-face interviews were found to be most suited for probing for answers, while at the same time, observing the behaviours of these individual participants. Designer diaries were used to record participants` self-reflection. These reflections provided rich data because of their descriptive language and detailed explanations, which made it easier for the researchers to read and understand aspects that had not explicitly surfaced through observations and interviews (Babapour, Rehammar & Rahe, 2012; Charlesworth, 2007).

2.2 Research Setup of the overall workshop and the design brief (tambah interview)

The workshop consisted of five sessions: S0, S1, S2, S3, and S4 (Figure 2). Sessions S1 through S3 addressed the creative development, where each session dealt with a specific design phase, but with a different design activity for each group. Session S0 was an introductory session, and session S4 was a final face-to-face interview.

Figure 2 Illustration of the workshop flow comprising five sessions

During session S0, participants were briefed on the processes and procedures as well as what they should expect from the study. The participants were then asked to record their background information and to sign a consent form agreeing to participate in the study. After the briefing session, each participant was given a design diary where they had to document their feelings and thoughts (i) before, (ii) during, and (iii) after the session regarding the product they created, as well as the processes they experienced. The design brief used for the purpose of this study was as follows:

STOOL: FURNITURE SOLUTIONS TO FIT CRAMPED SPACE IN URBAN CITY

LIVING

With the vast development and increase in low- and medium-cost apartment housing in Kuala Lumpur, (Malaysia´s capital), cities have become denser, and we all find ourselves wishing we had more space at

times, or want to make better use of the space we have. To facilitate space optimisation in small-space living, flexible and adaptable furniture, there is a need to develop furniture for multiple uses.

As a sub-section of this overarching design brief, you are tasked to design a stool for a small-space indoor living environment.

Although the complexity of the object (stool) was considered low-moderate in this design experiment, the researchers received initial feedback from the participants that the object was appropriate, because they are familiar with using it in their everyday life. Moreover, from a professional perspective, most participants have experienced designing a similar type of object, which is manageable in terms of size and complexity, provides ample opportunities for creative exploration in terms of form, structure, construction and functionality, and justifies a fair and balanced use of the three design tools: sketches prototypes and CAD models.

Sessions S1 through S3 were conducted to explicitly answer the research question. Three groups were guided in the following design activities: Sketching, Prototyping and CAD modelling (Howard et al., 2008). However, each group was subjected to different sequences of these design activities, as shown in Figure 2.

To further explore and confirm the findings, face-to-face interviews were conducted in session S4 with all participants recording their experiences regarding the process used. These interviews led to the emergence of the following sub-questions: (i) What expectations and preconceptions did the participants have regarding this design process? (ii) Which processes enhanced their imagination and stimulated their creativity? (iii) How satisfied were they with the overall design process? When answering these sub-questions, the participants were specifically asked to reflect on their processes, specifically those pertaining to critical thinking and creativity. All face-to-face interviews were recorded and transcribed, including the scripted questions. Each transcript was then analysed and cross-compared.

2.3 Selection of participants

81 professional designers with an affinity toward 3-D model making and sketching voluntarily took part in the experiment. The professional designers had 1 to 15 (mean = 7.22, st.dev = 2.77) years of experience. The gender breakdown of the participants was 57 males and 24 females. Finally, their ages ranged from 19 to 39 (mean = 22, st.dev. 5.17).

2.4 Workspace, facilities, and materials

The design workshop was conducted at the Industrial Design Department at the Universiti Teknologi MARA, Malaysia. Four design studios were made available for this workshop, where each of the participants had their own separate individual workspace.

As shown in Table 1, a diverse range of materials and aids were provided to facilitate the idea developed throughout the different stages. These materials and aids were chosen based on designers´ familiarity and frequency of use. Table 1 gives an overview of the physical aids and materials to facilitate the three design activities.

Table 1 Complete list of materials and aids provided during the experiment

Design Activities	Aids	Materials
Sketching	Pencil, pen, marker and pastel	A4 & A3 sketch paper
Prototyping	Cutter, scissors, masking tape, glue, wire cutter, cutting board	Boxboard, mounting board, rope, modelling clay, bamboo sticks, styrofoam, straw, wire, chenille stems, sponge, foam paper, balsa wood ,corrugated paper, EVA foam
CAD	Participants were asked to bring their own laptop	

2.5 Quantitative analysis methods

After the experiment, the number of ideas from all participants were counted and compared with the following independent variables: design activity, round number, group number (essentially the order of the design activity), and professional background. ANOVA tests were used when investigating the first three categories, followed by a Tukey test of Honest Significant Difference (Tukey's HSD) to see which categories significantly varied from the others. Pairwise post-hoc testing was used to determine whether there is a difference between the mean of all possible pairs using a studentized range distribution. This method tests every possible pair of all groups. Initially, the Tukey test was called the 'Honestly significant difference' test, or simply the 'T-test,' because this method was based on the t-distribution. The statistical assumptions of ANOVA should be applied to the Tukey method, as well. The Tukey test is a generous method to detect the difference during pairwise comparisons (less conservative).

1.6 Qualitative analysis methods

The design diaries were collected and analysed in this study. Each design diary was transcribed and reviewed using a data analysis approach suggested by Miles and Huberman (1994). The contents were then categorised to align them with the data from the interviews.

1.7 How expert reviews were used

With regards to assessing what is creative in a specific case (i.e. original and appropriate), few people can make this judgment. According to Amabile (1983), not many objective methods are available for evaluating creativity. Assessing whether an idea is creative or not was done through subjective expert evaluation (Shalley and Gilson, 2004)

Ideas were evaluated based on the quality, completeness and quantity of the representations by five Industrial Design experts (see Table 2). Quality and completeness of the ideas were assessed based on the following four criteria relative to the use of typical representations: (i) form and shape (ii) structure (iii) detailing and (iv) aesthetic appearance. These experts were asked to quantitatively evaluate the ideas on a 5-point Likert scale (1=Very poor; 2= Poor; 3=Average; 4=Good; 5=Excellent).

Table 2 Details of the experts

Expert Profile	Design Experience	Specialisation	Affiliation
Expert1: Researcher & Practitioner	12 years	Expert in product and sustainable design Research and design of healthcare products and services	Institute of Product Design, NTNU, Norway
Expert 2: Researcher, Practitioner & Educator	22 years	Expert in Furniture and Product Design, Research on Form giving and Meaning making	Institute of Product Design, NTNU, Norway
Expert 3: Practitioner & Educator	27 years	Expert in Furniture Design, research on applied aesthetics in design	Institute of Product Design, NTNU, Norway
Expert 4: Researcher, Practitioner & Educator	26 years	Expert in Design methodology and Industrial Design. Research on semantics and form giving	Industrial Design Department, University Technology MARA, UiTM, Malaysia.
Expert 5: Researcher, Practitioner & Educator	22 years	Expert in Product Design. Research on meaning making and creation.	Industrial Design Department, University of Lund, Sweden

Findings

This section presents the findings of the study in three parts. First, results are presented to provide the reader with an overview of the quantity of ideas from each of the three groups. Second, ideas produced for each design activity and design sessions were further dissected. Finally, the behaviours, attitudes, and mindsets of the designers were identified through analysing their design diaries.

3.1 Overview of the quantity of ideas from each of the three groups

Variations according to the order of design activities and tool uses showed interesting findings concerning participants´ attitudes and motivations. Figure 3 shows examples of sketches, CAD and prototypes models created with low and high levels of detailing according to the extent of completeness. Based on expert evaluations, only ideas marked as “high completeness” as shown in Figure 3 were counted whereas those marked as “low completeness” were not counted as an idea.

Figure 3 Example of the level of completeness for each representation. Low fidelity is illustrated in the pictures on the right, and high- fidelity ideas are illustrated on the left

Table 3 summarises the numbers of ideas from each group. The number of ideas converged (83 – 63 – 30) as Group 1 - participants applied design tools, which were typically expected to complement a classical design process. Group 3 (started with CAD) only produced a total of 135 ideas. Group 2 participants used prototypes in the early idea stage and produced a higher number of early ideas (112), compared to those groups that used sketches (Group 1) and CAD (Group 3). A similar pattern can be identified for the “idea development” stage, but the quantitative differences were not so significant. Interestingly, group 2 also produced the highest number of ideas through sketching during the “Final ideation” stage. This may be seen as a response to the discrepancy between sessions 1 and 2, where opportunities were created through prototyping in the “Early-ideation” stage, but not exploited through CAD in the “Idea development” stage. The experience of group 3 suggests the restrictive nature of CAD in the early idea stage as it did not stimulate subsequent ideations in sessions 2 and 3.

When comparing design tools within each stage, prototyping is most effective in generating early-stage ideas, whereas sketching due to certain circumstances may contribute significantly to final ideation sessions. In the “Idea development” stage, differences among activities / tool use are present but not significant with respect to generating quantity in idea development.

In total, 199 sketches, 124 CAD models and 211 physical prototypes were developed, which indicates that prototyping and sketching were considered the most effective ideation tools.

Table 3 The number of ideas produced by each group of each session

Group	SESSION 1 EARLY-STAGE IDEA		SESSION 2 IDEA DEVELOPMENT		SESSION 3 FINAL IDEA		OVERALL TOTAL IDEAS
	activities	total ideas	activities	total ideas	activities	total ideas	
Group 1	Sketching	83	Prototyping	63	CAD	30	176
Group 2	Prototyping	112	CAD	47	Sketching	64	223
Group 3	CAD	47	Sketching	52	Prototyping	36	135

As mentioned in the theoretical section, the design process alternates between divergent and convergent moves, while the overall trend is converging towards a limited number of final options. Hence, to test whether the participants created fewer ideas at the end of the process compared to the beginning, connections between the number of ideas and the sessions were explored. Here, a statistical difference was found between the number of ideas created during the different sessions ($F = 8.217$, $p = 0.00043$). Tukey's HSD test showed that the difference was significant between sessions S1 and S2 and between sessions S1 and S3 (Table 4). Participants came up with significantly more ideas in session S1 than in the following session.

Table 4 Results of Tukey's HSD test between the number of ideas and session number

	Difference	Sig.	Lower Bound	Upper Bound
S1 vs. S2	-.4815*	.022*	-.9061	-.0569
S1 vs. S3	.2593	.316	-.1654	.6839
S2 vs. S3	.7407*	.000***	.3161	1.1654

This finding confirmed the researchers´ assumptions, which indicated that the participants developed ideas according to the suggested design process, focusing on quantity through exploration and divergence during the first phase, and gradually converging more extensively during the detail development stages.

3.2 Flow of design activities

3.2.1 Design activities: Sketching - Prototyping - CAD

This group, who followed the traditional design process of sketching-prototyping-CAD (Figure 4), indicated their satisfaction when using sketches during the early design creation stages. The sketches in session S1 were mainly drawings of the concepts to be further developed in session S2 through prototyping, whereas CAD have been used in session S3 for presentation purposes only. Statistical analysis confirmed that there was a significant difference between the groups ($F = 6.866$, $P<0.001$), which will be explained when comparing with groups 2 and 3.

Conversely, when investigating the results from Tukey's HSD test (Table 5), it was found that the participants in group 2 starting with prototyping generated significantly more ideas than the participants of the other two groups (group 1 and 3). There were no significant differences between groups 1 and 3. This indicates that the group starting with sketching and CAD did not significantly differed in their early design ideas output. Hence, the constraining aspects of CAD observed during the qualitative observations cannot be confirmed statistically.

Table 5 Results from Tukey's HSD test comparing the number of ideas generated in the different groups (**p<0.1, *p<0.05)

	Difference	Sig.	Lower Bound	Upper Bound
Group 1 and 2	-.4815*	.022	-.9061	-.0569
Group 1 and 3	.2593	.316	-.1654	.6839
Group 2 and 3	.7407*	.000	.3161	1.1654

3.2.2 Design activities: Prototyping – CAD - Sketching

In group 2, prototyping has been used to develop and reflect upon initial ideas, which were refined in CAD. This refinement supposed to initiate a new round of creative ideation, which were addressed through sketches. Group 2 participants using prototypes during session S1 developed more varied ideas compared to Group 1, who started with conventional sketches. They constructed more, explored more, and were more hands-on engaged with " the materials". An example is shown in Figure 5.

Based on the measurements of divergence by Yoruk and Runco (2014), the ideas developed through prototyping showed higher flexibility and resolution. This finding is supported by the statistical analysis as shown in Table 6, where participants developed significantly more ideas when prototyping during session S1 ($p = 0.022$ and $p = 0.000$).

It is also worthwhile pointing out that materials used when prototyping seemed to inspire greater associations and further idea development as evidenced in the number of ideas being develop in the final ideation stages, even though the solution space was constraint in the "Idea development " stage because of CAD.

Table 6 Results from Tukey's HSD test of the number of ideas developed for each design activity and concerning to the sessions (*=p<0.1, **=p<0.05, ***=p<0.01)

Session 1	Difference	Sig.	Lower Bound	Upper Bound
Prototype vs Sketches	.4815*	.022**	.0569	.9061
Prototype vs. CAD	.7407*	.000***	.3161	1.1654
Sketches vs. CAD	.2593	.316	-.1654	.6839

3.2.3 Design activities: CAD – Sketching-Prototyping

In group, 3, CAD has been used to develop the building blocks for initial idea development, which were supposed to be further elaborated through sketches and refined through prototyping. Results showed a

statistically significant difference (p = .316 and p = 0.000) appearance between CAD and sketches, and CAD vs. prototype (Table 7), which indicates that participants generated significantly fewer ideas when using CAD. This suggests the constraining nature of CAD as design activities were aimed to generate a more specific solution in the early idea stages.

Table 7 Results from Tukey HSD test between the number of ideas and design activity

	difference	Sig.	Lower Bound	Upper Bound
CAD vs Sketches	-.2593	.316	-.6839	.1654
CAD vs Prototype	-.7407*	.000	-1.1654	-.3161
Sketches vs Prototype	-.4815*	.022	-.9061	-.0569

Based on their diary entries and general feedback sessions, participants in this group significantly mentioned their dissatisfaction with the process as compared to the other two groups (p = 0.000). Moreover, when prototypes was used in session S3, it only served to finalise the replication of ideas developed earlier (Figure 6). This was particularly observed in group 3, where half of the participants were not effective in exploring creative ideas in session S2 after using CAD in session S1. Hence, we can see that these participants failed to move beyond the solution space defined in session S1, because of the constraints imposed by CAD on creative exploration. This implies that despite significant advancements in CAD, designers still preferred traditional design tools, such as pens and pencils, particularly in the more conceptual stages of the design process, where incomplete collections of specifications and abstract design ideas drive creative exploration.

3.3 *Designers´behaviours, attitudes, and mindsets*

Complementary to S1 – S3 experiments, diary reflections showed that a majority of the participants, 75.3% (61) agreed that the prototyping process enhanced their imagination and creativity. As such, it can be confirmed that physical prototypes were most effective in creatively generating design ideas in the subsequent sessions. Furthermore, one professional designer during the interview described his design experience as follows:

“I started with modelling and then converted to CAD and final ideas using sketches, which was an easy start for me. At first, I was a bit sceptical about the process. Then I realized that with the prototype I could analyse the form and function of each development. I finalised my ideas very quickly; I could envision the structured and final form for development. This method is excellent, by starting with a prototype I can see the exact design problem…it helped me use my experiences and convert them into the design.”

Nevertheless, through interviews and diary entries, the participants unanimously agreed that CAD was most suitable to be used during the final detail development to confirm the shape and design of the product (p = 0.0000). A professional designer also added the following:

“In the first stages, it is better to use manual tools because we can produce ideas very fast, and can develop more ideas compared to digitally… When using CAD, there are so many limitations; for instance, you cannot use a certain shape such as organic and geometric shapes… it takes time to produce a shape…you would need a very skilled person to come up with good ideas.”

Participants starting with CAD significantly mentioned their dissatisfaction because of its restrictive explorative and creative capabilities, particularly when a broad variety of ideas need to be generated given a limited timeframe. Furthermore, implicit requirements for mastering CAD programs were found to be very demanding.

A majority of Group 1 and Group 3 participants used prototyping materials they were most familiar with. However, Group 2 participants were more receptive to explore and to be inspired by new and alternative materials. During the “Idea development” stage, a significant number of new ideas were made of prototypes using low-tech materials, such as clay, styrofoam and sticks, mainly to explore the physical form.

Discussion And Further Research

The aim of this study was to gain a better understanding of how the flow of design activities in the early stages of the engineering design process influence creativity in terms of securing divergence and defining solution spaces.

Prototypes were more effective than sketches and CAD in supporting divergent thinking as well as defining and broadening the solution space in the “early idea” stages. Prototypes also contributed most significantly to idea development. Compared to sketches, they showed an equal number of ideas, but more concrete, higher quality explorations. Participants using CAD for “early idea “(Group 3) were least explorative, somewhat frustrated and showed no signs of further explorations in the follow-up stages.

Interestingly, it has also been observed that the sequence of activities in the design process influenced the number of developed ideas. This calls for a conscious intervention on the usage of different design activities supporting the aims of a particular stage. From these findings, it can be postulated that the question now perhaps changes from which design activity to use, to when to use it. From this perspective, our findings indicated the benefit of not applying CAD modelling too early because doing so can limit exploration. This is also evident in the design industry, where two-thirds of design activities were supported by physical prototyping, whereas CAD modelling was only applied during the last one-third of the design process (Jensen, Lozano & Steinert, 2016).

Concerning further research, this study proposes the following two directions:

- To investigate additional permutations of design activities, such as “Prototyping - Sketching - CAD modelling”

- To extend prototyping in the early stages of the design process by engaging in hands-on design thinking. For example, future experiments can be conducted to find out how participants can jump directly, without sketching anything, to building with their hands whatever comes to their minds.

Furthermore, since material research for prototyping showed that the type and resolution of the materials provided in early prototyping determined the solution space for subsequent creative exploration (Jensen et al., 2016), further research could also explore the relationship between prototyping materials and resolution of ideas . At one end of the prototyping spectrum, the use of “Prototyping Toolkit Materials” allowed participants to express their ideas in both concrete and abstract ways (Lee and Park, 2021). However, these pre-made building blocks may to some extent restrict creative explorations. For example, participants may be too preoccupied with “what” could be created with the toolkit, rather than solving the brief. At the other end of the spectrum, all kinds of materials in their basic form open up vast opportunities for developing creative solutions. However, non-design students and authorities may have difficulties expressing their thoughts on these traditional, basic materials, because they are unfamiliar with them.

To further research on prototyping materials, it can also be interesting to investigate how the moderate low-fidelity nature of combining pre-defined building blocks (such as LEGO) in conjunction with other basic, raw materials such as cardboard, sticks and styrofoam should promote “Hands-on Thinking and Visualisation through Prototyping” in the early design stages. LEGO´s visualisation capabilities, when complemented with other materials have helped designers to explore, evaluate, modify and validate design ideas and solutions three-dimensionally (Isa and Liem, 2020; Ranscombe et al, 2020). LEGO´s low-fidelity properties can be easily iterated upon, without needing to deform its elements, due to its versatile construction and mechanism. In complementary use with LEGO, the other basic materials are easily mouldable and transformable when generating multiple and challenging free-form shapes, as well as easily represented in a rough and incomplete format due to the characteristics of the material. The flexible use of these combined tools in early development stages managed to reveal design problems more accurately for more reflective, iterative, and interpretative solution finding (Isa and Liem, 2020; Ranscombe et al, 2020). Moreover, the low-fidelity characteristics of LEGO facilitate increased discussions, when working in teams, because of its ambiguity.

In terms of implications for design education, this study may influence the teaching of design methodology and reasoning to be more versatile. Besides guiding students through a structured and positivistic process of “Analysis – Syntheses” (Bamford, 2002), introducing different sequences of design activities offers alternative pathways for design ideation. In particular, focusing on prototyping as a tool for research and development in the early stages motivates conjectural moves in finding design solutions through more constructive, reflective (Schøn, 1987) and hermeneutic reasoning (Snodgrass and Coyne, 1996). Therefore, the co-existence of structured problem solving, reflective practice and hermeneutic approaches may facilitate the completeness, scope, resolution and quality of the ideas.

Conclusion

This study has outlined the interplay among three design activities: Prototyping, Sketching, and CAD modelling. This paper suggests that the sequence of design activities in the engineering design process determines how effective these design tools are in facilitating early idea, idea development and final idea.

Findings indicate that starting with prototyping in the early idea stage not only contributed significantly to a higher quality of ideation, but also to a broader degree of exploration. Moreover, using CAD-modelling too early in the engineering design process might constrain idea generation and prematurely limit the exploration of the solution space. Furthermore, this paper stressed the need for a balanced approach towards early prototyping to facilitate creativity through alternating concrete as well as abstract moves in the generation of design solutions. This means that sufficient room should be created for those engaged in early prototyping to work with predefined toolkits, more basic materials as well as elements and sub-assemblies. Given a context, where pre-defined elements and basic/raw materials can be applied simultaneously, development of concrete design solutions may be facilitated while allowing “ambiguity” to remain. In other words, high-fidelity and low-fidelity characteristics should be balanced to provide clear visual explanations of the design intent.

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Figures

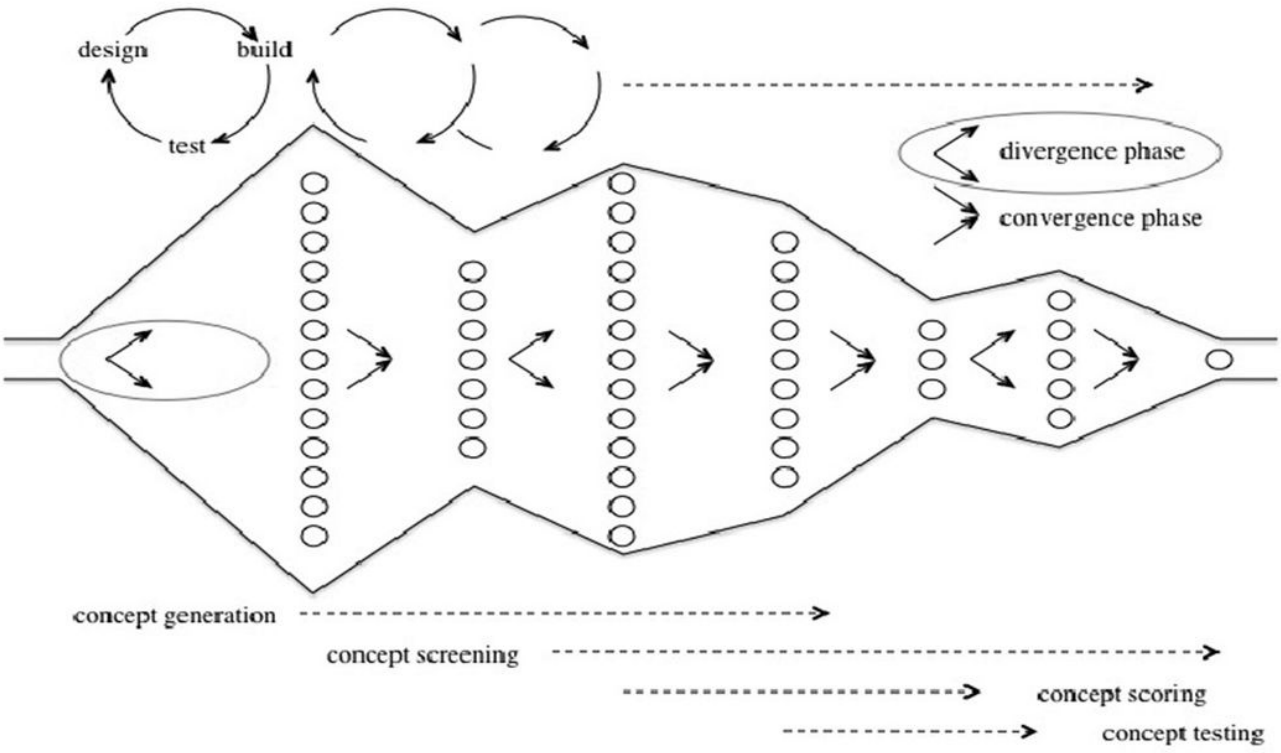


Figure 1

Design process as an iteration of divergence and convergence steps or prototype cycles, adapted from Leifer and Steinert (2012)

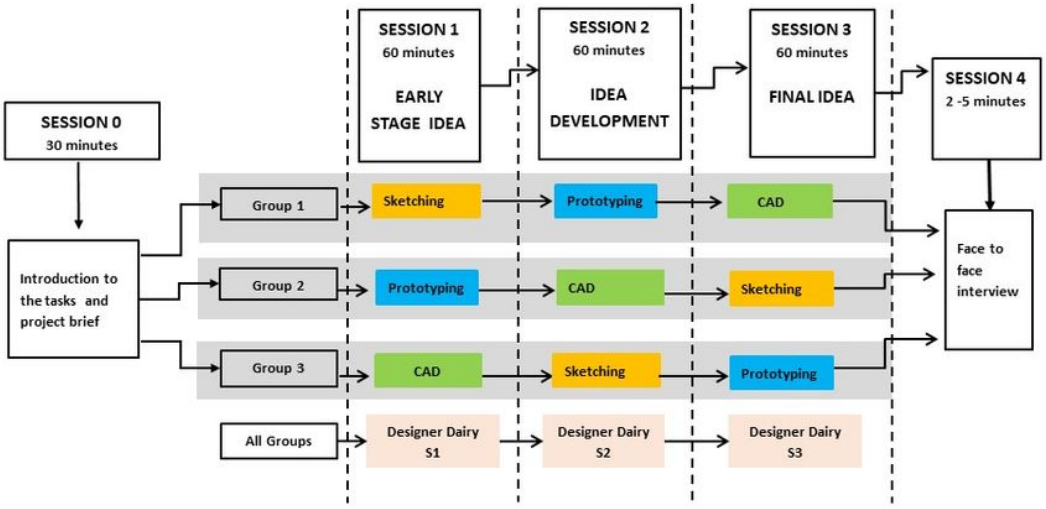


Figure 2

Illustration of the workshop flow comprising five sessions

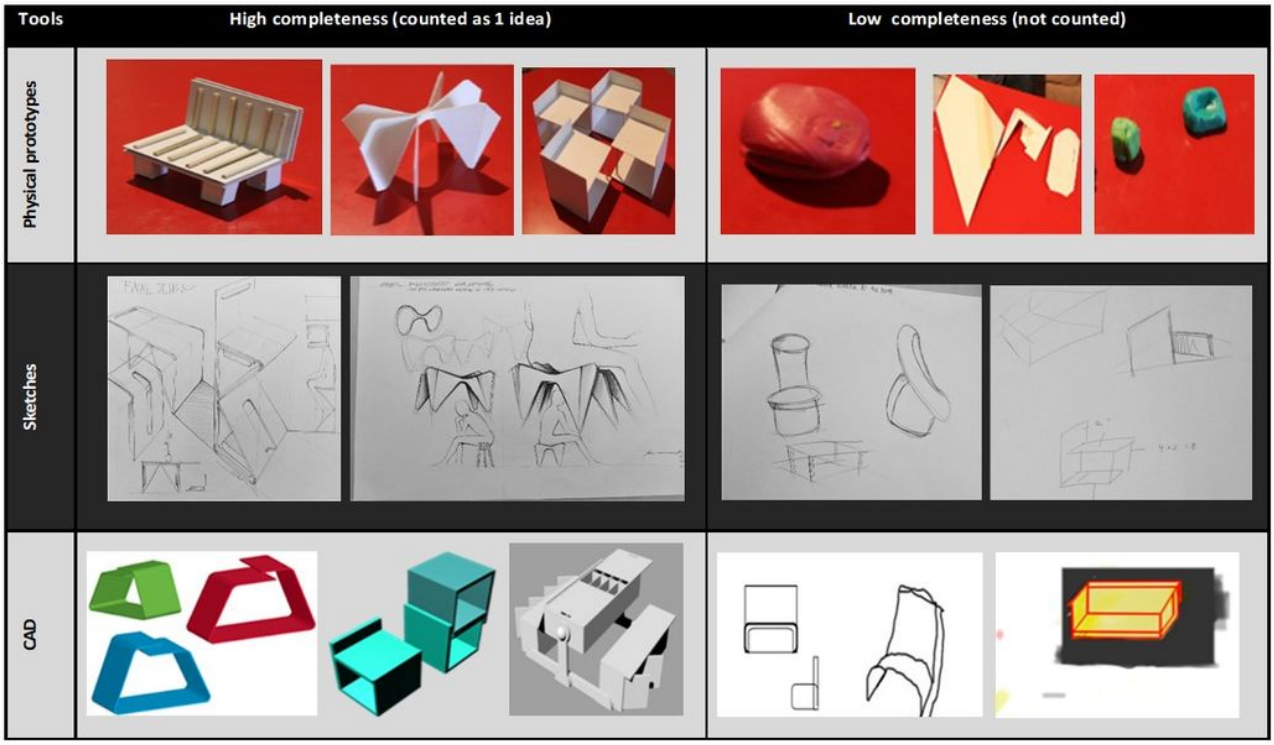


Figure 3

Example of the level of completeness for each representation. Low fidelity is illustrated in the pictures on the right, and high- fidelity ideas are illustrated on the left

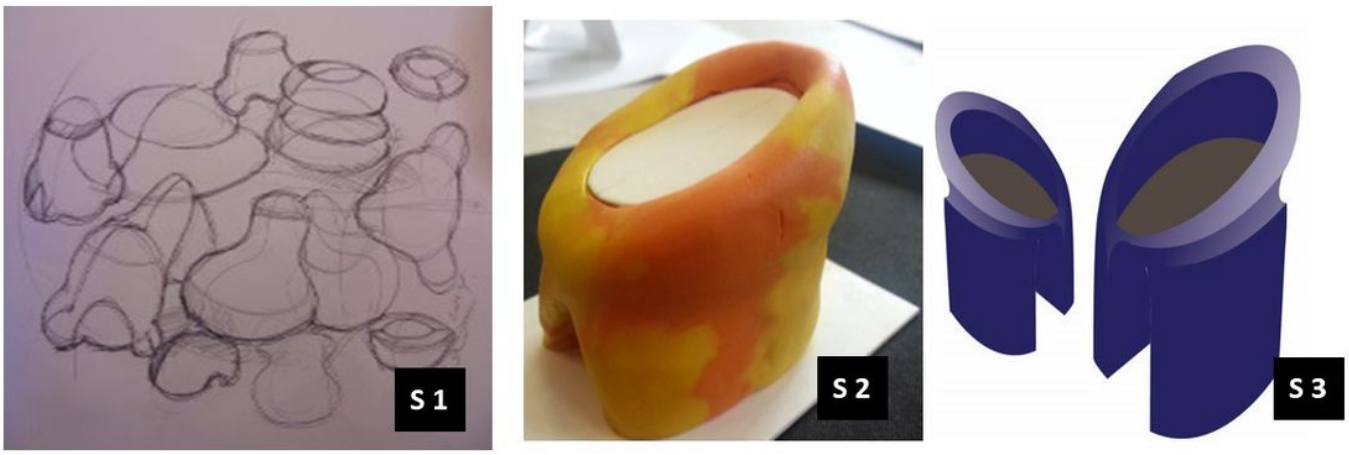


Figure 4

Example of ideas developed during S1 and the following ideas developed in S2 and S3

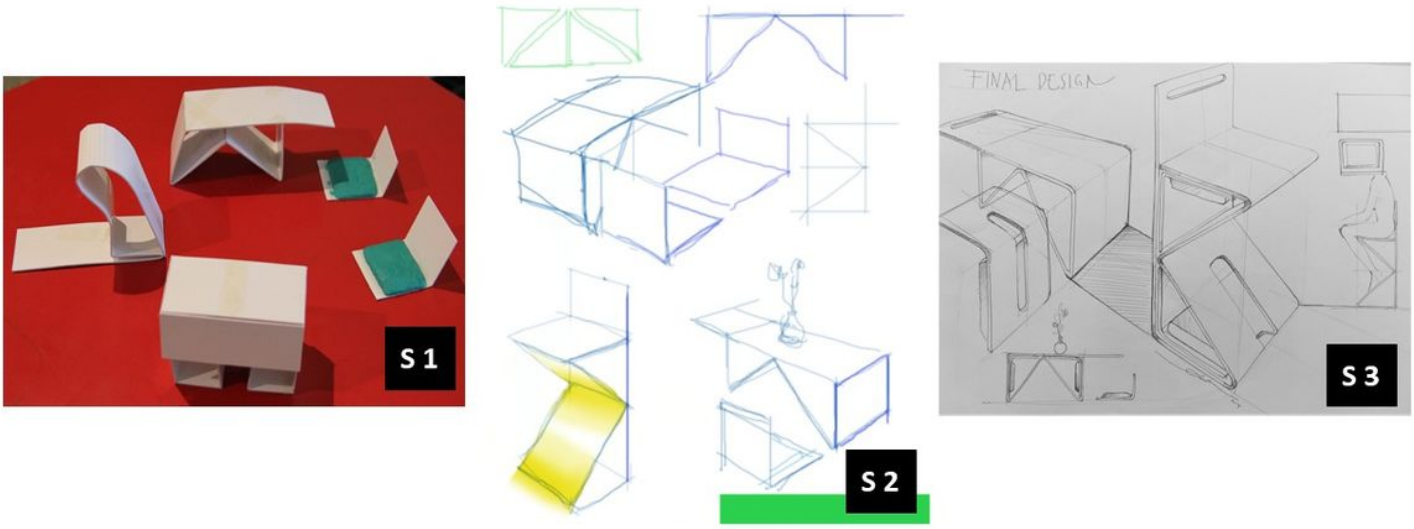


Figure 5

Example of ideas developed during S1 and the following ideas developed in S2 and S3

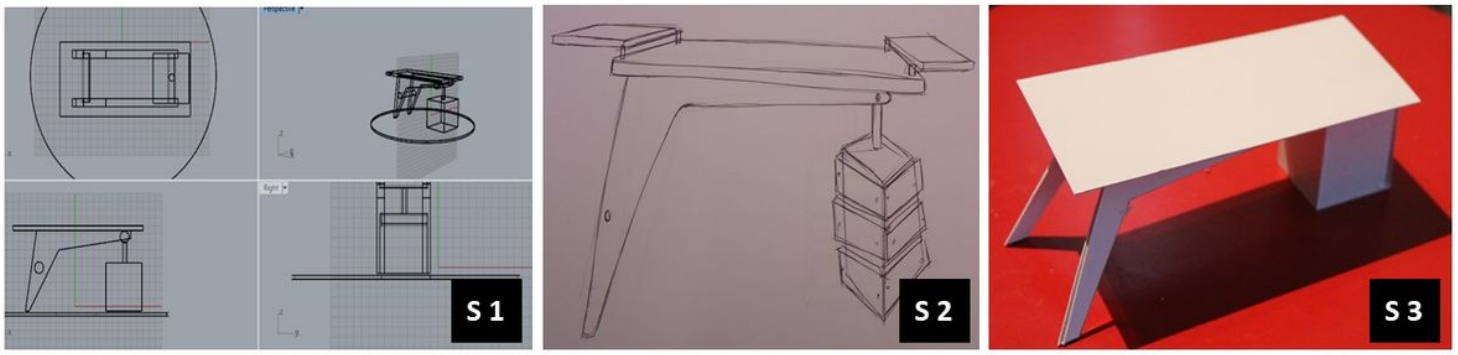


Figure 6

Example of idea development from Group 3. It should be noted how the idea in S1 is replicated, but not explored in S2 and S3